

CODE-LAYOUT, READABILITY AND REUSABILITY

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized into templates, static, models, notebooks, and dataset folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names such as temperature, humidity, prediction, and scaled_data are used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like home(), predict_page(), and prediction() are clearly named.
5	Code Comments	Inline and block comments explained	Yes	Major sections and logic are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Project is divided into Flask routes, templates, CSS, and machine learning modules.

7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code has been removed. Duplicate and unnecessary code has been removed.
8	Error Handling	Exceptions and errors are handled	Yes	Uses <code>try-except</code> blocks to handle prediction and input-related errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Function	Python	Flask Prediction Route	High
2	StandardScaler	Scikit-learn	Input data Preprocessing	High
3	XGBoost Model	XGBoost	Flood Prediction	High
4	HTML Form Template	HTML/CSS	Prediction Page	Medium
5	CSS Styling	CSS	Home & Prediction Pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with separate folders for templates, static files, models, and dataset.
Readability	5	Meaningful variable names, proper formatting, and easy-to-understand code.
Reusability	5	Modular design using reusable Flask routes, model loading, and CSS components.
Documentation / Comments	5	Code includes comments explaining major sections and workflow.
Overall Score	4.8	Clean, modular, maintainable, and deployment-ready project.